

From One Chart to a Full Financial Model

How the Rent vs. Buy Calculator Evolved

Every capability on the following slides started as a question — not a spec.

Where We Started

Most rent-vs-buy calculators only compare monthly payments. Build one that also asks what the down payment could have earned elsewhere.

- A 30-year net worth comparison, renter vs. buyer
- One clear answer: the **break-even year**

Hyper-Local Rent vs. Buy Opportunity Cost Calculator

Most calculators ignore the opportunity cost of your down payment. This one invests it in the stock market and tracks a true 30-year net worth comparison.

[Pitch Deck](#) [Download PDF Report](#) [Share Scenario](#) [Save as My Defaults](#) [Reset to Defaults](#)

THE PROPERTY

Property Address

Zip Code (California)

Home Price

Monthly Rent (comparable)

HOMEOWNERSHIP COSTS

Down Payment

Mortgage Rate

Property Tax Rate

BREAK-EVEN POINT

Year 15

Buying overtakes renting's investment returns in year 15.

Mortgage (P&I) / mo

\$3,287

Buyer Net Worth (Yr 30)

\$1,520,423

Renter Net Worth (Yr 30)

\$1,550,453

30-Year Net Worth Projection

The chart displays two lines representing net worth over 30 years. The 'Buyer Home Equity' line (green) starts at \$0 and rises to approximately \$550,000 by Year 15, then continues to rise more steeply. The 'Renter's Net Worth' line (blue) starts at approximately \$130,000 and rises steadily to approximately \$1,550,000 by Year 30. A vertical dashed line at Year 15 indicates the break-even point where the Buyer's net worth surpasses the Renter's net worth.

THE QUESTION

"How can we make this feel local, not generic?"

Zip-code-aware defaults

Type a zip code, get real market numbers instantly — home price, rent, and property tax rate — instead of guessing at three unfamiliar sliders.

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THE PROPERTY

Property Address

e.g. 123 Main St, San Jose, CA 95123

Zip Code (California)

94110

Loaded defaults for San Francisco (Mission)

Import from Zillow Screenshot

Home Price

\$1.4M

Monthly Rent (comparable)

\$3.4K

HOMEOWNERSHIP COSTS

Down Payment

20% (\$270K)

Mortgage Rate

6.5%

Property Tax Rate

1.18%



THE QUESTION

"What costs are we missing?"

Insurance and maintenance, modeled properly

Home insurance and maintenance don't inflate at the same pace as rent, so they got their own sliders and their own inflation rates — not lumped together as one vague "cost of owning" number.

The section was renamed from "Mortgage & Costs" to "**Homeownership Costs**" to match what it actually covered by this point.

THE QUESTION

"A single guess isn't good enough — what about risk?"

A second view: Monte Carlo simulation

Markets don't move in a straight line. A toggleable second view runs 500 simulated 30-year outcomes and shows a range, not false precision.

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THE PROPERTY

Property Address
e.g. 123 Main St, San Jose, CA 95123

Zip Code (California)
e.g. 94110

Import from Zillow Screenshot

Home Price
\$650K

Monthly Rent (comparable)
\$2.8K

HOMEOWNERSHIP COSTS

Down Payment
20% (\$130K)

Mortgage Rate
6.5%

Property Tax Rate
1.10%

BREAK-EVEN POINT (MEDIAN)

Year 10

In the median simulated scenario, buying overtakes renting in year 10.

Mortgage (P&I) / mo
\$3,287

Buyer Net Worth (Yr 30), median
\$1,479,682

Renter Net Worth (Yr 30), median
\$1,115,245

30-Year Net Worth Projection

Deterministic Monte Carlo

Buying wins in 64% of 500 simulated 30-year scenarios. Shaded bands show the 10th-90th percentile range. Each simulated year draws one real historical calendar year (1987-2026) and applies that same year's Stock Market Return and home-price change together — so a simulated 2008 pairs the real stock/bond/gold return with the real Case-Shiller home-price drop from 2008, keeping downturns that historically hit both housing and investments together paired. Both are re-centered to your Stock Market Return and home appreciation sliders respectively.

THE QUESTION

"Can that simulation be grounded in real history, not a guess?"

Real market history, not invented statistics

The simulation originally sampled from a made-up statistical curve. It now resamples from what actually happened:

- Real S&P 500 annual returns, **1928–2025**
- The real Case-Shiller national home price index, **1987–2025**

Every simulated year is a year that genuinely occurred, just reshuffled.

THE QUESTION

"What about taxes — isn't that the biggest lever?"

Modeling what actually shows up on a tax return

- Mortgage interest and property tax deductions, weighed against the standard deduction
- Capital gains tax on the renter's investment gains
- The home-sale tax exclusion buyers get — \$250K or \$500K tax-free — which most calculators ignore entirely

THE QUESTION

"Taxes got complicated — can we make sense of it?"

Giving taxes their own home

Every tax-related input moved into one dedicated section. A state tax rate was added too, since California taxes investment gains as ordinary income — another detail most calculators miss.

THE QUESTION

"How do I share exactly what I'm looking at?"

One click, one link

Every input and every setting — encoded into a single URL. No accounts, no saved files, no re-explaining your numbers to someone else.

THE QUESTION

"Do I have to type in every number by hand?"

Upload a screenshot instead

Drop in a screenshot of a listing and the key numbers get pulled out automatically. Nothing is applied silently — every extracted value is shown for review and confirmation first.

THE QUESTION

"Can I take this analysis with me?"

A real report, one click away

Your exact scenario — every chart, every input, every assumption — turns into a clean, shareable document instead of living only on screen.

THE QUESTION

"What if I don't want to bet on the stock market?"

Choose your own investment

Stocks, Treasury bonds, or gold — each with its own real historical data behind it. Same rigor, whichever path you're actually weighing.

THE QUESTION

"How do we know a change didn't quietly break the math?"

A real test suite, not just eyeballing the chart

Every core calculation — mortgage amortization, tax logic, Monte Carlo percentiles, zip lookup, screenshot parsing — got automated tests, run in CI before every deploy. One of those tests caught a real currency-formatting bug that only showed up on the CI server's Node version, before it ever reached production.

THE QUESTION

"What if a crash year for stocks landed on a calm year for housing in the same simulated trial?"

Correlated, not just random

The Monte Carlo simulation used to draw the stock return and the home-price change for a given year **independently** — so a simulated trial could pair 2008's stock crash with a historically strong housing year, which never actually happened. Now each simulated year draws **one** real historical year and applies that year's return and home-price change together, so downturns that hit both markets at once stay paired.

THE QUESTION

"Can I set my own starting point, and label which property this actually is?"

Your own defaults, and a way back to them

"Save as My Defaults" remembers your own baseline assumptions in the browser; "Reset to Defaults" snaps everything back to it (or the factory scenario, if you've never saved one) in one click. A Property Address field labels exactly which listing a scenario is for, on screen and on the PDF report.

THE QUESTION

"Can I use this without the web app at all?"

The same model, as a spreadsheet

A Google Apps Script builds a live-formula Google Sheet replica — the same deterministic engine and the same 500-trial correlated Monte Carlo simulation, driven entirely by your own inputs, auditable cell by cell.

Every Feature Started With a Question

None of this was planned upfront. Each capability came from asking "can we also..." — and the model just kept getting more honest.

onevvish.github.io/rentVsbuy